

ASAL ROODBARI

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

- **Ph.D., Machine Learning — ISyE**

2024 – Present

- Focus: Machine learning for temporal representation learning and survival analysis, with applications in healthcare
- GPA: 3.7/4.0

Sharif University of Technology

Tehran, Iran

- **B.Sc., Electrical Engineering — Control & Dynamical Systems**

2020 – 2024

National Organization for Development of Exceptional Talents (NODET)

Sari, Iran

- **Higher Secondary Education, Mathematics & Physics**

2014 – 2020

HONORS & AWARDS

- **Machine Learning for Personalized Risk Assessment in Idiopathic Scoliosis**
with Shriners Hospitals for Children

- **Poster Award — 2nd Place** 2026
State of Science Meeting, Shriners Children's Hospital
- **Poster Award — 1st Place** 2025
NSF AIVO Summit for AI Institutes Leadership
- **John H. Moe Award — Best Basic Science/Translational Research Poster** 2025
Scoliosis Research Society (SRS) Annual Meeting
- **Nominated — Best Translational Research Oral Presentation** 2025
Pediatric Orthopaedic Society of North America (POSNA) Annual Meeting

PUBLICATIONS

Journals

1. A. Roudbari, S. Sabzekar, A. Dehghani, A. Safaeiestalkhzir, and Z. Amini. “*The Impact of Network Indices Integration on Traffic Flow Imputation Accuracy: A Machine Learning Approach.*” *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 4, pp. 5411–5421, 2025. [pdf]
2. A. Roudbari, M. Khosravy and O. Witkowski. “*Genetic Algorithm Applications to Power Management.*” *Frontiers in Genetic Algorithm Theory and Applications*, Springer Nature, Singapore, 2024. [pdf]

Conferences

1. M. C. Welborn, A. Roudbari, R. Zandehshahvar, J. O. Sanders, P. Van Hentenryck, A. F. Samdani, J. D. Stone, R. H. Cho, S. C. Poon, V. R. Talwalkar, M. Szczodry, P. Gupta, and S. Sienko. “*Biomarker Longitudinal Idiopathic Scoliosis Study: The Impact of Growth on Curve Progression in Braced Patients with Idiopathic Scoliosis Who Experience Curve Progression vs. Curve Regression.*”

Pediatric Orthopaedic Society of North America (POSNA) Annual Meeting, 2025. Nominated for Best Translational Research Oral Presentation.

2. M. C. Welborn, A. Roudbari, R. Zandehshahvar, P. Van Hentenryck, A. F. Samdani, R. H. Cho, J. D. Stone, S. C. Poon, V. R. Talwalkar, M. Szczodry, P. Gupta, J. O. Sanders, and S. Sienko. “*Collagen X Biomarker (CXM) is Predictive of Growth Cessation in Idiopathic Scoliosis.*” *Scoliosis Research Society (SRS) Annual Meeting, 2025.* Awarded John H. Moe Award for Best Basic Science/Translational Research Poster. [pdf]
3. A. Roudbari, M. Dalvand, N. Pasiar, N. Asadi, M. Yousefi, and M. T. Masouleh. “*Autonomous Vision-Based Robotic Grasping of Household Objects: A Practical Case Study.*” *Proceedings of the 2023 11th RSI International Conference on Robotics and Mechatronics (ICRoM), IEEE, 2023.* [pdf]

RESEARCH EXPERIENCE

- **AI for Scoliosis** Sep. 2024 – Present
Supervisors: Dr. MohammadReza Zandehshahvar, Prof. Pascal Van Hentenryck AI4OPT, Georgia Institute of Technology & Shriners Hospital
 - Developed a personalized machine learning pipeline for predicting growth cessation in Adolescent Idiopathic Scoliosis (AIS) using irregular longitudinal patient data.
 - Collaborated with orthopedic clinicians at Shriners Hospital to align model interpretation with real-world clinical decision-making.
- **Line Graph Neural Network for Spatial Traffic Flow Imputation** Jul. 2023 – Dec. 2023
Supervisor: Prof. Zahra Amini Sharif University of Technology, Tehran, Iran
- **Object Grasping — 2-Finger Gripper and Delta Robot (Bachelor’s Project)** Apr. 2023 – Sep. 2023
Lab: Human and Robot Interaction Lab, University of Tehran
Supervisor: Prof. Mehdi Taale Masouleh
- **Applications of Genetic Algorithm in Power Management** Feb. 2023 – Apr. 2023
Lab: Cross Labs (Remote, Japan)
Supervisor: Dr. Mahdi Khosravy
- **Object Localization at Sea: Estimation Methods in Wide Ranges** Jul. 2022 – Oct. 2022
Lab: Marine Technology, Risk and Intelligence in Marine Systems, Aalto University
Supervisors: Prof. Osiris AV Banda, Dr. Meriam Chaal

TECHNICAL SKILLS

- **Programming & ML Frameworks:** Python (PyTorch, scikit-learn, NumPy, Pandas, OpenCV), R, MATLAB

LEADERSHIP & SERVICE

- **Iranian Student Association (ISA), Georgia Institute of Technology**
President Oct. 2025 – Mar. 2026
Communications Director Jan. 2025 – Oct. 2025
Served as President following a tenure as Communications Director. Directed all communication channels (email, social media, events), coordinated cultural and academic programming, enhanced outreach materials, and sustained active engagement across the Iranian student community.
- **Student Leadership Member, AI4OPT** Mar. 2025 – Present
Georgia Institute of Technology
Represented PhD student interests in seminars, mentorship initiatives, and research culture discussions. Organized social events for the institute and participated in Seth Bonder Camp.

- Communications Director, Resana (EE Student Council) May 2021 – Sep. 2022
 Sharif University of Technology, Iran
 Led communication strategy and outreach for the Electrical Engineering Department’s student council; organized multiple academic seminars, competitions, student workshops, voluntary charity initiatives, and student well-being support activities.
- Competition Organizer and Communication Director, “Spark” (I.D.E.A. Association) Spring 2022
 Sharif University of Technology, Iran
 I.D.E.A. Association, Germany
 Co-organized an international student competition; oversaw participant communications and provided technical support throughout the event.
- Event Host & Organizer — Emeet1, Emeet2, ReACT 2021 – 2022
 Sharif University of Technology, Iran
 Hosted university-wide cultural events with 300+ participants; managed logistics and outreach communication, promotion, and on-site coordination.